Industrial Entomology & Aquaculture

Class 10 (Technical)

Unit – 2

Introduction of cocoon, cocoon quality & its Characteristics

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Introduction of Cocoon

- 5th instar stage larva gradually stops feeding & slowly it release proteins (Fibroin & Sericin) from its mouth in the form of saliva, which become thread when it comes in contact with the air. The larva uses this thread to slowly wrap itself around, keeping itself at the center. This structure which has pupa in the middle and the thread on the outer region is called cocoon.
- Cocoon color is varies as per Spp. Of Silkworm







Quality of Cocoon

- It refers to the measure of excellence or a state of being free from diseases, defects and abnormalities.
- Superior quality can be obtained by strict and consistent following of measurable & verifiable standards.
- Cocoons output should always be of reeling quality otherwise it will be rejected in the market.
- The quality of cocoon is measured by various characteristics such as shell percentage, defective cocoon %, average filament length, average non-broken filament length, reeling % etc.







- The quality of cocoons may vary even in the same batch due to differences in quality of feeding-leaf & bed-spacing.
- Univoltine & Bivoltine cocoons are superior quality compared to the Multivoltine.







Major factor affecting the quality of cocoons

A. Temperature, RH & Aeration during mounting:

Ideal temperature of rearing house to get good quality cocoons with high reelability is 25°C -26°C, & RH 65-75%.

B. Mountage / Mounting device:

There are various types of mounting device are available but, rotary mounting frames can provide good ventilation, which will result in improved quality & reelability of cocoons.







Classification of Cocoons

Good Cocoon

- Uniform Cocoon in shape & color
- Easy to extract filament
- High shell %
- Uniformly wrapped
- Good reelability



Bad Cocoon

- Irregular shape
- Cut or hole in the cocoon
- Discolored or bad color (spotted color)
- Less shell %
- Thin end cocoon or un-uniformly wrapped



Characteristics of Cocoon

1. **Color:** it is temporary and washed away with washing of sericin. It is vary depending upon the race of silkworm. (Yellow, White, Grey etc.)
2. **Shape**: it also varies according to the species of silkworm, it is heavily affected by the mounting process. Useful in identifying the race (Chinese = Elliptical/Egg Shaped, Japanese = Peanut Shell Shape)
3. **Size:** Varies according to race of silkworm, rearing season & harvesting condition.
4. **Raw Silk %:** It varies from 65-84% for weight of the cocoon shell & 12-20% for weight of the whole fresh cocoon.
5. **Filament Length:** 1 cocoon = 600 to 1500 m filament of which 80% is reelable.
6. **Weight:** Weight of cocoons of pure breeds ranges from 2.2-1.5g, while hybrid = 2.5-1.8g per cocoon.

Any Query ??

Thank You !!

See You In Next Class ……..